Result 1 (exact maps vanish on the image). [Mathlib exact pairs, Function.Exact.apply_apply_eq_zero]
Let K be a type. Let *field* be a field structure on K. Let M, N and P be elements of Mod_K. Let f be a linear map from M to N. Let g be a linear map from N to P. Let x be an element of M. Assume $\mathrm{exact}(f, g)$ (`exactness of f and g`). Then

$$g(f(x)) = 0.$$

Result 2 (exact sequence interior at each position). [Topology foundations, exact sequence definition]
Let K be a type. Let *field* be a field structure on K. Let S be a sequence of modules over K. Let n be a natural number. Assume interior exactness of S (`exactness`). Then

$$\mathrm{exact}_{n+1}(S).$$

LLM-assisted definitions and supporting proofs (gpt-6, 2026-11-10). Interior exactness concerns positive degrees; no degree-zero endpoint condition or Mayer–Vietoris theorem is asserted.